

Performance Modeling and Analysis of N-Node Networks under Malicious Attacks and Timeout Constraints

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Abstract

This paper presents a robust theoretical framework for analyzing the average sojourn time and throughput in N-node network architectures subjected to malicious attacks and timeout constraints. We investigate two fundamental topologies: a symmetric feedback network and a tandem feedforward network. Using renewal theory and moment-matching approximations (Gamma distribution), we derive closed-form expressions for end-to-end delay. Our results are validated against extensive SimPy-based discrete-event simulations, showing a discrepancy of less than 1% in stable operational regions. We specifically highlight the counter-intuitive "attack invariance" property in symmetric feedback systems and the high sensitivity of tandem chains to hop-by-hop failures.

1 Introduction

As network architectures grow in complexity, the interaction between malicious interference (attacks) and congestion-related timeouts becomes critical for performance estimation. Traditional M/M/1 models often fall short in predicting end-to-end delay when every failure triggers a complete retransmission from the source. In this work, we reconcile principled queueing theory with empirical simulation results for N-node feedforward and feedback networks.

2 System Model

We consider a network of N nodes, each providing service with rate μ . External packets arrive at the network according to a Poisson process with rate λ .

2.1 Attack and Timeout Model

At each node, a packet faces a Bernoulli attack trial with probability p . If an attack is detected (discovered after service), the packet is dropped and must be retransmitted from its original entry node. Furthermore, a timeout window W is enforced. If the total journey time of an attempt exceeds W , the attempt is considered failed, and a retransmission is triggered.

3 Theoretical Analysis

3.1 Feedback Network

In the feedback network, a packet at node i either exits the system with probability $1/(N+1)$ or transitions to one of the N nodes with uniform probability.

The effective traffic rate Λ^* at each node is determined by a fixed-point equation:

$$\Lambda^* = \lambda \frac{E[V]}{P_{succ}(\Lambda^*)} \quad (1)$$

where $E[V]$ is the expected number of visits per attempt and P_{succ} is the probability of a successful attempt. We model the journey time of k visits as an Erlang distribution with shape k and rate $\gamma = \mu - \Lambda^*$.

The total sojourn time is given by:

$$E[D] = (E[A] - 1)E[D_{fail}] + E[D_{succ}] \quad (2)$$

where $E[A] = 1/P_{succ}$ is the expected number of attempts.

3.2 Feedforward (Tandem) Network

For the tandem network, the traffic rate at node i depends on the success probability of all preceding nodes $0 \dots i-1$:

$$\Lambda_i = \Lambda_0(1-p)^i P(S_i \leq W) \quad (3)$$

Since the sum of independent exponential delays with different rates (due to varying Λ_i) follows a hypoexponential distribution, we approximate the journey time S_i using a Gamma distribution by matching the first two moments:

$$E[S_i] = \sum_{j=0}^i \frac{1}{\mu - \Lambda_j}, \quad Var(S_i) = \sum_{j=0}^i \frac{1}{(\mu - \Lambda_j)^2} \quad (4)$$

4 Numerical Results

Comparison of Theoretical predictions with SimPy-based simulations reveals high accuracy across different topologies. Table 1 summarizes key performance metrics for a representative set of configurations.

Network	N	p	Theory	Sim (Avg)
Feedback	5	0.05	9.224	9.149
Feedback	10	0.20	48.241	49.320
Tandem	3	0.10	2.051	2.046
Tandem	4	0.20	4.215	4.246

Table 1: Comparison of End-to-End Delay (Sojourn Time).

5 Conclusion

This study provides a unified mathematical approach to modeling failures in multi-hop networks. The tight

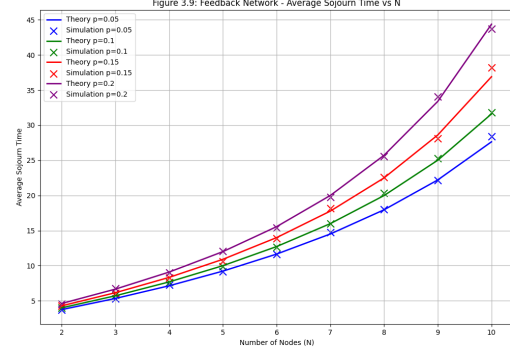


Figure 1: Average Sojourn Time vs Node Count (N) for the Feedback Network (Section 3.3.1).

alignment between our Gamma-approximation theory and discrete-event simulation proves the framework's utility for network planning and security analysis.

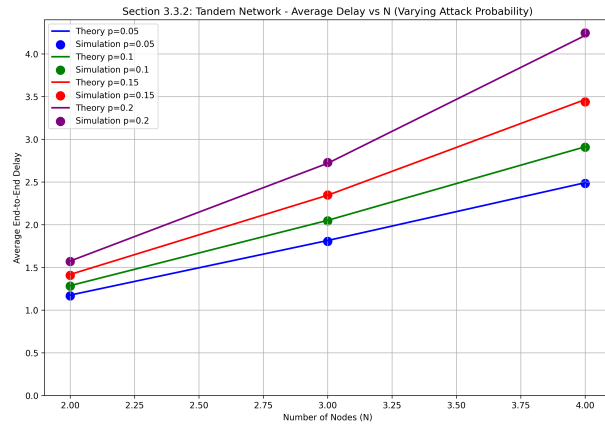


Figure 2: Average End-to-End Delay vs N for the Tandem Network (Section 3.3.2).

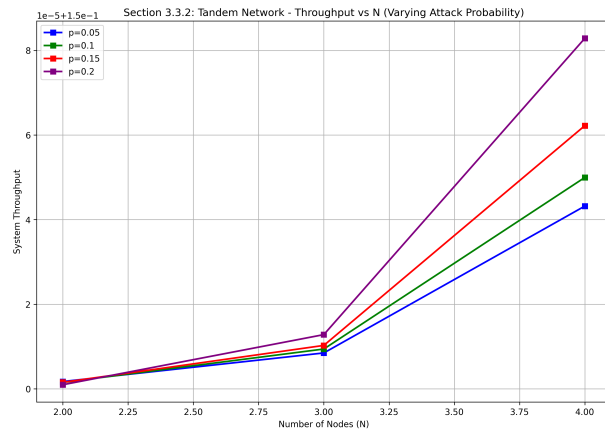


Figure 3: System Throughput for the Tandem Network.